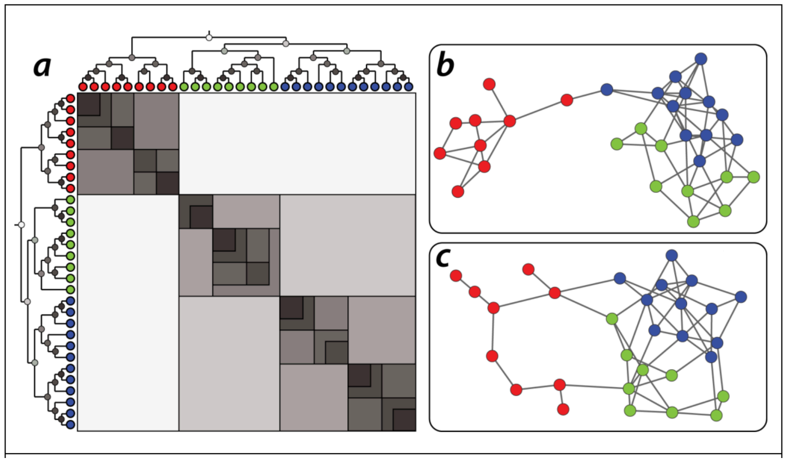
L12: Hierarchical Graph Model Applications



One application of the HRG model is to generate a large ensemble of networks – all of them following the same probabilistic model estimated based on a single given network.

These networks can then be used in simulation studies. For instance, imagine testing a new routing algorithm on many different “Internet topologies” that have been generated from a single snapshot of the current Internet topology.

Another application may be that this ensemble of networks is used as the ”null model” in testing whether a given network follows a certain model or not. For instance, suppose that we have generated an HRG model based on a sample of healthy brain networks – and we have also generated an HRG model based on a sample of brain networks from schizophrenia patients. We can then use these two models to test whether a given brain network belongs in one or the other group.

The following visualization shows two different networks (b and c) that were both generated using the HRG model shown at the left (a). The model has 30 nodes. Note that the shading of the internal dendrogram nodes is such that darker means higher probability values. Together with the dendrogram, this visualization also shows the corresponding adjacency matrix, Note that the blue and green communities belong to a larger community that is more loosely connected – and they are less likely to be connected to the red community.

Another application of the HRG model is to detect the presence of *false-positive* or *false-negative* edges. This is important in practice because the measured network data is often incorrect or incomplete – especially in the context of biology of neuroscience. So, suppose that we are given a single network G and we construct an HRG model $\tilde{G}$ based on G.

If $\tilde{G}$ predicts that there is a high probability that two nodes *X* and *Y* are connected, while these two nodes are not connected in G, we may have a missing edge (false negative) in G.

On the other hand, if $\tilde{G}$ predicts that there is a low probability that two nodes *X* and *Y* are connected, and these two nodes are connected in G, we may have a spurious edge (false positive) in G.

Food For Thought

Can you think of other applications of the HRG model?